Lecture - 11

$$T : \underbrace{(V, +, \cdot, \mathbb{R})} \longrightarrow \underbrace{(W, +, \cdot, \mathbb{R})}$$

$$\boxed{f : \text{Set} \longrightarrow \text{Set}}$$

$$T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ y+z \end{bmatrix}$$

$$(\mathbb{R}^3, +, \cdot, \mathbb{R}) \qquad (\mathbb{R}^2, +, \cdot, \mathbb{R})$$

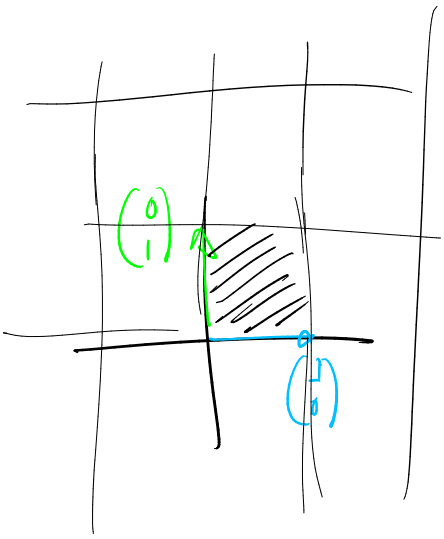
From now, we are interested in Linear Transformation where input and output space are same.

Operator

For example take,

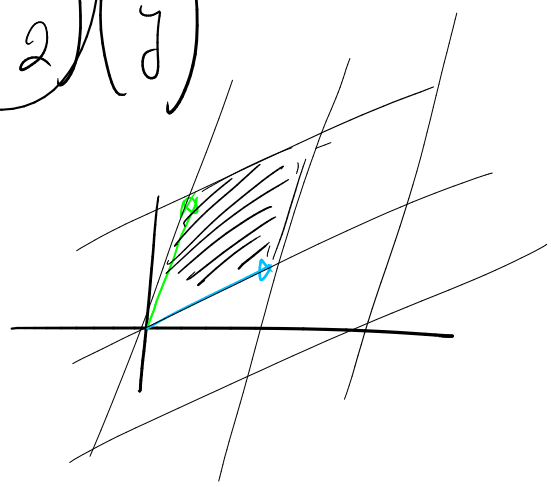
$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x+y \\ x+2y \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$



$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$



$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

So, we are interested to get such input vectors, who doesn't change the direction on the output space.

$\vec{v}$, $\quad T(\vec{v}) = \boxed{A\vec{v} = \lambda \vec{v}}$

$\Rightarrow$ v doesn't change the direction but it can change the length

$$A\vec{v} = \lambda \vec{v}$$

$$\Rightarrow A\vec{v} - \lambda I \vec{v} = 0$$

$$\Rightarrow \boxed{(A - \lambda I)}\vec{v} = 0 \longrightarrow \text{vector}$$

Matrix input = Zero vector.

<u>Non-trivial</u> solution,

$$\underbrace{\det(A - \lambda I) = 0}_{} \longrightarrow \text{find } \lambda$$

$\Rightarrow$ that matrix will be dependent

$\Rightarrow$ free variable available.

$\Rightarrow$ Non-trivial solution

Example:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}_{2\times 2} \quad \left(\text{square matrix}\right)$$

$$\det\left(A - \lambda I\right) = 0$$

$$\Rightarrow \det\left[A - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2\times 2}\right] = 0$$

$$\Rightarrow \det\left[\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right] = 0$$

$$\Rightarrow \det\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = 0$$

$$\Rightarrow (2-\lambda)^2 - 1 = 0$$

$$\Rightarrow 4 - 4\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 3) = 0$$

$$\Rightarrow \boxed{\lambda = 1, \quad \lambda = 3}$$

In this direction length won't be changed

length will be three

$T : \mathbb{R}^2 \to \mathbb{R}^2$

Step-2

$$(A - \lambda I)\vec{v} = \vec{0}$$

consider, $\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\left[\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right] \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

For $\lambda = 1$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & \boxed{0} & 0 \end{array}\right)$$

$x + y = 0 \Rightarrow \boxed{x = -y}$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ y \end{pmatrix} = y \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

For $\lambda = 3$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 1 & -1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & \boxed{0} & 0 \end{array}\right)$$

$$-x + y = 0 \Rightarrow x = y \qquad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ y \end{pmatrix} = y \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A\vec{v} = \lambda \vec{v}$$

direction doesn't change = Eigenvector

How much the length change = Eigen value.

Example

$$A = \begin{pmatrix} 2 & -3 & 6 \\ 0 & 5 & -6 \\ 0 & 1 & 0 \end{pmatrix}$$

$\det(A - \lambda I) = 0$

$$\det \begin{pmatrix} 2-\lambda & -3 & 6 \\ 0 & 5-\lambda & -6 \\ 0 & 1 & -\lambda \end{pmatrix} = 0$$

$$\Rightarrow (2-\lambda)(-5\lambda + \lambda^2 + 6) = 0$$

$\lambda = 2,$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - 2\lambda + 6 = 0$$

$$\Rightarrow \lambda(\lambda - 3) - 2(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 3, \quad \lambda = 2$$

step-2

$$(A - \lambda I)\vec{v} = \vec{0} \qquad \vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{pmatrix} 2-\lambda & -3 & 6 \\ 0 & 5-\lambda & -6 \\ 0 & 1 & -\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

For $\lambda = 2$

$$\begin{pmatrix} 0 & -3 & 6 \\ 0 & 3 & -6 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 0 & -3 & 6 & 0 \\ 0 & 3 & -6 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 0 & -3 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 0 & -1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} \boxed{0} & -1 & 2 & 0 \\ 0 & \boxed{0} & 0 & 0 \\ 0 & 0 & \boxed{0} & 0 \end{array}\right)$$

$$-y+2z=0 \qquad y=-2z$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ -2z \\ z \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + z\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$$

$\lambda=3$ Try yourself

Similar matrix

Q: find the diagonalization of A?

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \qquad B = ? \text{ (Diagonal)}$$

find

Eigenvalues and Eigenvectors

$\lambda = 1$, eigenvector, $\vec{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$\lambda = 3$, eigenvector, $\vec{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Put the eigenvector in the columns

$$P = \begin{array}{cc} \text{eigenvector} & \text{eigenvector} \\ \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \end{array} \qquad P = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$P^{-1}AP = \frac{1}{-2}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{array}{c} \lambda=1 \\ \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \\ \lambda=3 \end{array} = B$$

$$\tilde{P}^{-1}A\tilde{P} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

Q: Find A^{10000}

$$\boxed{A = P^{-1}BP} \Longrightarrow A^{10000} = \underbrace{(P^{-1}B\cancel{P})}\times\underbrace{(\cancel{P^{-1}}B\cancel{P})}\times\cdots$$

$$= P^{-1}B^{10000}P$$

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}^{10000} = \begin{pmatrix} a^{10000} & 0 \\ 0 & b^{10000} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{10000} = P^{-1}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}^{10000}P$$

$$= \begin{pmatrix} & \\ & \end{pmatrix}\begin{pmatrix} 3^{10000} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} & \\ & \end{pmatrix}$$

length, inner product

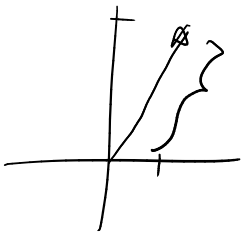
$$(V, \oplus, \odot, \mathbb{R})$$

addition,

multiplication

length

$$2i+3j \longmapsto \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$



$$\sqrt{2^2+3^2}$$

$$(1i+2j)\cdot(2i+3j) = (1\cdot 2)+(2\cdot 3) = 8$$

(Generalize the dot product)

$$\underbrace{u\cdot v} = (\text{formula}) = \mathbb{R}$$

$$(V, \oplus, \odot, \mathbb{R})$$

$$B : V \times V \longmapsto \mathbb{R}$$

(inner product) $\langle u, v \rangle \longmapsto \mathbb{R}$.

① Linear in first argument

(i) $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$

(ii) $\langle \alpha u, w\rangle = \alpha \langle u, w\rangle$

② Symmetric

$$\langle u, v\rangle = \langle v, u\rangle$$

③ $\langle u, u\rangle \geq 0$

④ $\langle 0, u\rangle = 0$ (non-degenerate)

$$u \cdot v = \boxed{u_x \cdot v_x + u_y \cdot v_y + u_z \cdot v_z}$$

① $\langle A, B\rangle = tr(A^T B)$ Matrix space

$\langle f, g\rangle = \int_{-1}^{1} f(t)\, g(t)\, dt$ function space

$$\left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \right\rangle = \operatorname{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^T \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \right)$$

$$= \operatorname{tr}\left(\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}} \right)$$

$$= \operatorname{tr} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

$$= 5.$$